Vaxcel®

CEFTRIAXONE INJECTION

COMPOSITION:

Each vial contains
Ceftriaxone Sodium equivalent to
Ceftriaxone 500mg and 1g.

PRESENTATION:

White to yellowish - orange crystalline powder.

INDICATIONS:

Infections caused by pathogen sensitive of Ceftriaxone:

- Sepsis;
- Meningitis;
- Disseminated Lyme borreliosis (early and late stages of the disease);
- Abdominal infections (peritonitis, infections of the biliary and gastrointestinal tracts);
- Infections of the bones, joints, soft tissue, skin and of wounds;
- Infections in patients with impaired defence mechanisms;
- Renal and urinary tract infections;
- Respiratory tract infections, particularly pneumonia, and ear, nose and throat infections;
- Genital infections, including gonorrhea. Perioperative prophylaxis of infections.

PHARMACOLOGY:

Mechanism of Action:
Ceftriaxone is a third generation cephalosporin antibiotic with an extended spectrum of activity and increased potency against Gram-negative bacteria including the Enterobacteriaceae, *Haemophilus influenzae, Moraxella (Branhamella) catarrhalis*, and Neisseria spp., but only moderate activity against Pseudomonas spp. The bactericidal activity of Ceftriaxone results from inhibition of bacterial cell wall synthesis. It is highly stable to most beta-lactamases, both penicillinases and cephalosporinases, of gram-positive and gram-negative bacteria. Ceftriaxone is usually active against the following microorganisms in vitro and in clinical infections:

Gram negative aerobes:
Acinetobacter lwoffii, Acinetobacter anitratus (mostly A. baumannii), Aeromonas hydrophila, Alcaligenes faecalis, Alcaligenes odorans, Caligenes-like bacteria, Borrelia burgdorferi, Capnocytophaga spp., Citrobacter diversus (including C.amalonaticus), Citrobacter freundii*, Escherichia coli, Enterobacter aerogenes*, Enterobacter cloacae*, Enterobacter spp. (other)*, Haemophilus ducreyi, Haemophilus influenzae, Haemophilus parainfluenzae, Hafnia alvei, Klebsiella oxytoca, Klebsiella pneumoniae**, Moraxella catarrhalis (former Branhamella catarrhalis), Moraxella osloensis, Moraxella spp. (other), Morganella morganii, Neisseria gonorrhoea, Neisseria meningitidis, Pasteurella multocida, Plesiomonas shigelloides, Proteus mirabilis, Proteus penneri*, Proteus vulgaris, Pseudomonas fluorescens*, Pseudomonas spp. (other)*, Providencia rettgeri*, Providencia spp. (other) Salmonella typhi, Salmonella spp. (non-typhoid), Serratia marcescens*, Serratia spp. (other), Shigella spp., Vibrio spp., Yersinia enterocolitica, Yersinia spp. (other).* * some isolates of these species are resistant to Ceftriaxone, mainly due to the production of the chromosomally encoded beta-lactamase.
**some isolates of these species are resistant due to production of extended spectrum, plasmid-mediated beta-lactamase.
NOTE: Many strains of the above microorganisms that are multiply resistant to other antibiotics, e.g. aminopenicillins and ureidopenicillins, older cephalosporins and aminoglycosides, are susceptible to Ceftriaxone. *Treponema pallidum* is sensitive in vitro and in animal experiments. Clinical investigations indicate that primary and secondary syphilis respond well to Ceftriaxone therapy. With a few exceptions clinical *P. aeruginosa* isolates are resistant to Ceftriaxone.

Gram positive aerobes:
Staphylococcus aureus (methicillin-sensitive), Staphylococci coagulase-negative, *Streptococcus pyogenes* (β-hemolytic, group A), *Streptococcus agalactiae* (β-hemolytic, group B), β-hemolytic Streptococci (non-group A or B), *Streptococcus viridans, Streptococcus pneumoniae.*
NOTE: Methicillin-resistant Staphylococcus spp. are resistant to cephalosporins, including Ceftriaxone. In general, *Enterococcus faecalis, Enterococcus faecium* and *Listeria monocytogenes* are resistant.

Anaerobic organisms:
Bacteroides spp. (bile-sensitive)*, Clostridium spp. (excluding *C.difficile*), *Fusobacterium nucleatum,* Fusobacterium spp. (other), *Gaffkya anaerobica* (formerly *Peptococcus*), Peptostreptococcus spp.
* some isolates of these species are resistant to Ceftriaxone due to β-lactamase-production.
Note: Many strains of β-lactamase-producing Bacteroides spp. (notably *B.fragilis*) are resistant. *Clostridium difficile* is resistant. Susceptibility to Ceftriaxone can be determined by the disk diffusion test or by the agar or broth dilution test using standardized techniques for susceptibility test such as those recommended by NCCLS (National Committee for Clinical Laboratory Standards). The NCCLS issued the following interpretative break-points for Ceftriaxone:

	Susceptible	Moderately susceptible	Resistant
Dilution test inhibitory concentrations in mg/l	≤ 8	16-32	≥ 64
Diffusion test (disk with 30ug Ceftriaxone), inhibition zone diameter in mm	≥ 21	20-14	≤ 13

Microorganisms should be tested with the Ceftriaxone disk since it has been shown by in-vitro tests to be active against certain strains resistant to cephalosporin class disk.

Pharmacokinetics:
Ceftriaxone is a nonlinear dose-dependent pharmacokinetics because of its proteins binding.

Absorption:
The maximum plasma concentration after a single i.m. dose of 1g is about 81mg/l and is reached in 2-3 hours after administration. The area under the plasma concentration-time curve after i.m. administration is equivalent to that after i.v. administration of an equivalent dose, indicating 100% bioavailability of intramuscularly administered Ceftriaxone.

Distribution:
The volume of distribution of Ceftriaxone is 7-12l. Ceftriaxone has shown excellent tissue and body fluid penetration after a dose of 1-2g; concentrations well above the minimal inhibitory concentrations of most pathogens responsible for infection are detectable for more than 24 hours in over 60 tissues or body fluids including lung, heart, biliary tract / liver, tonsil middle ear and nasal mucosa, bone as well as cerebrospinal, pleural, prostatic and synovial fluids. On intravenous administration, Ceftriaxone diffuses rapidly into the interstitial fluid, where bactericidal concentrations against susceptible organisms are maintained for 24 hours.

Protein binding:
Ceftriaxone is reversibly bound to albumin, and the binding decreases with the increases in concentration, e.g. from 95% binding at plasma concentrations of <100mg/l to 85% binding at 300mg/l. Owing to the lower albumin content, the proportion of free Ceftriaxone in interstitial fluid is correspondingly higher than in plasma.

Penetration into particular tissues:
Ceftriaxone penetrates the inflamed meninges of neonates, infants and children: Ceftriaxone concentrations exceed 1.4 mg/l in the Cerebrospinal Fluid (CSF) 24 hours after i.v. injection of Ceftriaxone in doses of 50-100mg/kg (neonates and infants respectively). Peak concentration in CSF is reached about 4 hours after i.v. injection and gives an average value of 18mg/l. Mean CSF levels are 17% of plasma concentration in patients with bacterial meningitis and 4% in patients with aseptic meningitis. In adult with meningitis patients, administration of 50mg/kg leads within 2-24 hours to CSF concentration several times higher than the minimum inhibitory concentrations required for the most common meningitis pathogens. Ceftriaxone crosses the placental barrier and is excreted in the breast milk at low concentrations.

Metabolism:
Ceftriaxone is not metabolized systemically; but is converted to inactive metabolites by the gut flora.

Elimination:
Total plasma clearance is 10-22ml/min. Renal clearance is 5-12ml/min.
50-60% of Ceftriaxone is excreted unchanged in the urine, while 40-50% is excreted unchanged in the bile. The elimination half-life in adults is about 8 hours.

Pharmacokinetics in special clinical situations:
In neonates, urinary recovery accounts for about 70% of the dose. In infants aged less than 8 days and in elderly persons aged over 75 years the average elimination half-life is usually two to three times that in young adults. In patient with renal or hepatic dysfunction, the pharmacokinetics of Ceftriaxone are only minimally altered and the elimination half-life is only slightly increased. If kidney function alone is impaired, biliary elimination of Ceftriaxone is increased; if liver function alone is impaired, renal elimination is increased.

DOSAGE AND ADMINISTRATION:

Standard dosage:
Adult & children over 12 years: The usual dosage is 1-2g of Ceftriaxone once daily (every 24 hours). In severe cases or in infections caused by moderately sensitive organisms, the dosage may be raised to 4g, once daily.
Neonates, infants and children up to 12 years: The following dosage schedules are recommended for once daily administration:
Neonates (up to 14 days): 20-50mg/kg bodyweight once daily. The daily dose should not exceed 50mg/kg. It is not necessary to differentiate between premature and term infants.
Infants and children (15 days to 12 years): 20-80mg/kg once daily.
For children with bodyweights of 50kg or more, the usual adult dosage should be used.
Intravenous doses of ≥ 50mg/kg bodyweight should be given by infusion over at least 30 minutes.
Elderly patients: The dosages recommended for adults require no modification in geriatric patients.
Do not use diluents containing calcium, such as Ringer's Solution or Hartmann's Solution, to reconstitute Ceftriaxone. Particulate formation can result.

Duration of therapy:
The duration of therapy varies according to the course of the disease. As with antibiotic therapy in general, administration of Ceftriaxone should be continued for a minimum of 48-72 hours after the patients have become afebrile or evidence of bacterial eradication has been obtained.

Combination therapy:
Synergy between Ceftriaxone and aminoglycosides has been demonstrated with many gram-negative Bacteria under experimental conditions. Although enhanced activity of such combinations is not always predictable, it should be considered in severe, life threatening infections due to microorganisms such as *Pseudomonas aeruginosa*. Because of physical incompatibility the two drugs must be administered separately at the recommended dosages.

Special dosage instruction:
Meningitis: In bacterial meningitis in infants and children, treatment begins with doses of 100mg/kg (up to maximum of 4g) once daily. As soon as the causative organism has

been identified and its sensitivity determined, the dosage can be reduced accordingly. The following duration of therapy has shown to be effective:

Neisseria meningitis	4 days
Haemophilus influenzae	6 days
Streptococcus pneumoniae	7 days

Lyme borreliosis: 50mg/kg up to a maximum of 2g in children and adults, once daily for 14 days.
Gonorrhoea (penicillinase-producing and non penicillinase - producing strains): a single i.m. dose of 250mg.
Perioperative prophylaxis: A single dose of 1-2g depending on the risk of infection of 30-90 minutes prior to surgery. In colorectal surgery, administration of Ceftriaxone with or without a 5-nitro-imidazole, e.g. omidazole (separate administration, see Method of administration) has been proven effective.
Impaired renal and hepatic function: In patients with impaired renal function, there is no need to reduce the dosage of Ceftriaxone provided hepatic function is intact. Only in cases of preterminal renal failure (creatinine clearance < 10ml/min) should the Ceftriaxone dosage not exceed 2g daily. In patients with liver damage, there is no need for the dosage to be reduced provided renal function is intact.
In patients with both severe renal and hepatic dysfunction: the plasma concentrations of Ceftriaxone should be determined at regular intervals and if necessary the dose should be adjusted. In patients undergoing dialysis no additional supplementary dosing is required following the dialysis. Plasma concentrations should, however, be monitored, to determine whether dosage adjustments are necessary, since the elimination rate in these patients may be altered.

Method of administration:
As a general rule the solutions should be used immediately after preparation. Reconstituted solutions retain their physical and chemical stability for 24 hours at room temperature (or 72 hours in the refrigerator at 2-8^{o}C). The solutions range in color from pale yellow to amber, depending on the concentration and length of storage. The coloration of the solutions is of no significance for the efficacy or tolerance of the drug.

Intramuscular injection:
For i.m. injection, Ceftriaxone 250mg or 500mg is dissolved in 2ml, and Ceftriaxone 1g in 3.5ml, of 1% lidocaine hydrochloride solution and injected well within the body of relatively large muscle. It is recommended that not more than 1g be injected at one site. The lidocaine solution should never be administered intravenously.

Intravenous injection:
For i.v. injection, Ceftriaxone 250mg or 500mg is dissolved in 5ml, and Ceftriaxone 1g in 10ml sterile water for injections. The intravenous administration should be given over 2-4 minutes.

Intravenous infusion:
The infusion should be given over at least 30 minutes. For i.v. infusion, 2g Ceftriaxone is dissolved in 40ml of one of the following calcium-free infusion solutions: sodium chloride 0.9%, sodium chloride 0.45% + dextrose 2.5%, 5-10% dextrose, 6% dextran in 5% dextrose, hydroxy ethyl starch 6-10% water for injections. Ceftriaxone solutions should not be mixed with or piggy backed into solutions containing other antimicrobial drugs or into diluent solutions other than those listed above, owing to possible incompatibility.

CONTRAINDICATION:
Ceftriaxone is contraindicated in patients with known hypersensitivity to cephalosporin antibiotics. In patients hypersensitivity to penicillin, the possibility of allergic cross-reactions should be borne in mind. Ceftriaxone is contraindicated in neonates (≤ 28 days of age) if they require (or are expected to require) treatment with calcium - containing intravenous solutions, including calcium - containing infusions such as parenteral nutrition, because of the risk of precipitation of Ceftriaxone - Calcium.

PRECAUTIONS:
As with other cephalosporins, anaphylactic shock cannot be ruled out even if a thorough patient history is taken. Pseudomembranous colitis has been reported with nearly all antibacterial agents, including Ceftriaxone. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of antibacterial agents. Superinfections with non-susceptible microorganisms may occur as with other antibacterial agents. Shadows which have been mistaken for gallstones have been detected on sonograms of the gallbladder, usually following doses higher than the standard recommended dose.

Use in Pregnancy and Lactation:
Ceftriaxone crosses the placental barrier. Safety in human pregnancy has not been established. Reproductive studies in animals have shown no evidence of embryotoxicity, fetotoxicity, teratogenicity or adverse effects on male or female fertility, birth or perinatal and postnatal development. In primates, no embryotoxicity or teratogenicity has been observed. Low concentrations of Ceftriaxone excreted in human milk. Caution should be exercised when Ceftriaxone is administered to a nursing women.

Use in Peadiatric:
Studies have shown that Ceftriaxone, like some other cephalosporins, can displace bilirubin from serum albumin. Therefore, caution should be exercised when considering Cefriaxone treatment in hyperbilirubinemia neonates. Ceftriaxone should not be used in neonates (especially premature) at risk of developing bilirubin encephalopathy. During prolong treatment the blood profile should be checked at regular intervals.

WARNING:
Ceftriaxone must not be mixed or administered simultaneously with calcium-containing solutions or products, even via different infusion lines. Calcium containing solutions or products must not be administered within 48 hours of last administration of Ceftriaxone. Cases of fatal reactions with Calcium-Ceftriaxone precipitates in lungs and kidneys in both term and premature neonates have been described. In some cases the infusion lines and times of administration and calcium-containing solutions differed. In patients other than neonates, Ceftriaxone and Calcium - containing solutions may be administered sequentially to one another if the infusion lines are thoroughly flushed between infusions with a compatible fluid. Diluents containing calcium, such as Ringer's solution or Hartmann's solution, are not to be used to reconstitute Ceftriaxone vials or to further dilute a reconstituted vial for intravenous administration because precipitate can form. Ceftriaxone must not be administered simultaneously with calcium - containing intravenous solutions, including continuous calcium - containing infusions such as parenteral nutrition via a Y - site, because precipitation of Ceftriaxone - Calcium can occur. Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Ceftriaxone, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactams agents. If an allergic reaction occurs, Ceftriaxone must be discontinued immediately and appropriate alternative therapy instituted.

SIDE EFFECTS:
Systemic side effects:
Gastrointestinal complaints (about 2% of cases): loose stools or diarrhoea, nausea, vomiting, stomatitis and glossitis.
Hematological changes (about 2%): eosinophilia, leukopenia, granulocytopenia, hemolytic anemia, thrombocytopenia, isolated cases of agranulocytosis (<500 per mm^{3}) have been reported, most of them after 10 days treatment and following total doses of 20g or more.
Skin reaction (about 1%): exanthema, allergic dermatitis, pruritus, urticaria, edema, Isolated cases of severe cutaneous adverse reactions (erythema multiforme, Stevens-Johnson syndrome or Lyell's syndrome / toxic epidermal necrolysis) have been reported.
Other, rare side effects: headache and dizziness, symptomatic precipitation of Ceftriaxone Calcium salt in the gall bladder, increase in liver enzymes, oliguria, increase in serum creatinine, genital mycosis, fever, shivering and anaphylactic or anaphylactoid reactions.
Pseudomembranous enterocolitis and coagulation disorders have been reported as very rare side effects. Very rare cases of renal precipitation have been reported, mostly in children older than 3 years and who have been treated with either high daily doses (e.g. >80mg per kg/day) or total doses exceeding 10 grams and presenting other risk factors (e.g. fluid restriction, confinement to bed, etc.). This event may be symptomatic or asymptomatic, may lead to renal insufficiency, and is reversible upon discontinuation of Ceftriaxone.
Nervous system disorders:
Frequency 'not known': Encephalopathy*

*Reversible encephalopathy has been reported with the use of ceftriaxone, particularly when high doses are administered in patients with renal impairment and additional predisposing factors such as older age, pre-existing central nervous system disorders.

Local side effect:
In rare cases, phlebitic reactions occurred after i.v. administration. These may be minimized by slow (2-4 minutes) injection. Intramuscular injection without lidocaine solution is painful.

DRUG INTERACTIONS:
No impairment of renal function has been observed after concurrent administration of large doses of Ceftriaxone and potent diuretics (eg. Furosemide). There is no evidence that Ceftriaxone increases renal toxicity of aminoglycosides. No effect similar to that of disulfiram has been demonstrated after ingestion of alcohol subsequent to the administration of Ceftriaxone. Ceftriaxone does not contain an N-methylthiotetrazole moiety associated with possible ethanol intolerance and bleeding problems of certain other cephalosporins. The elimination of Ceftriaxone is not altered by probenecid.

OVERDOSAGE AND TREATMENT :
In the case of overdosage, drug concentration would not be reduced by hemodialysis or peritoneal dialysis. There is no specific antidote. Treatment of overdosage should be symptomatic.

SPECIAL REMARKS:
Incompatibilities:
Ceftriaxone should not be added to solutions containing calcium such as Hartmann's solution and Ringer's solution. Based on literature reports Ceftriaxone is incompatible with amsacrine, vancomycin and fluconazole and aminoglycosides.

Influence on diagnostic tests:
In patients treated with Ceftriaxone the Coombs' test may rarely become false-positive. Cefriaxone, like other antibiotics, may result in false-positive tests for galactosemia. Likewise, nonenzymatic methods for the glucose determination in urine may give false-positive results. For this reason, urine-glucose determination during the therapy with Ceftriaxone should be done enzymatically.

STORAGE:
Store below 30^{O}C. The freshly reconstituted solution is recommended. Its efficacy is maintained for at least 24 hours at room temperature (or 72 hours in the refrigerator at 2-8^{O}C).

KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK

PACK QUANTITIES:
Single vial per box & 10 vials per box.

Further information can be obtained from pharmacist, physician or the manufacturer.

Vaxcel Ceftriaxone 500mg Injection: MAL20014218AZ
Vaxcel Ceftriaxone 1g Injection: MAL20014220AZ

Product Registration Holder & Manufactured By:
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Date of Revision: 07/08/23